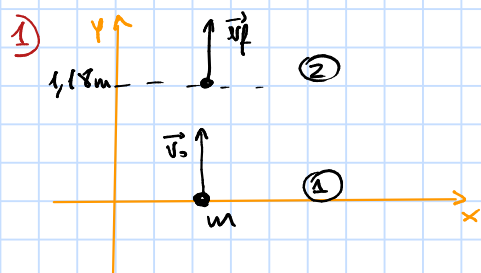


PER CAPIRE

TEOREMA DELLE FORZE VIVE



$$m = 0,120 \text{ kg}$$

$$v_0 = 7,85 \text{ m/s}$$

$$v_f = ?$$

All'andata $\rightarrow$ moto rettilineo uniformemente accelerato

$$\begin{cases} x(t) = v_0 t + \frac{1}{2} a t^2 + x_0 = v_0 t - \frac{1}{2} g t^2 \\ x(t) = v_f t + \frac{1}{2} a t^2 + x_0 = v_f t - \frac{1}{2} g t^2 + 1,18 \end{cases} \rightarrow$$

con $x(t)$ e lo spazio percorso

$$1,18 = 7,85t - \frac{1}{2} g t^2$$

$$v_0 t - \frac{1}{2} g t^2 = v_f t - \frac{1}{2} g t^2 + 1,18$$

$$7,85t - \frac{1}{2}(10)t^2 = v_f t - \frac{1}{2}(10)t^2 + 1,18$$

$$7,85t - 5t^2 = v_f t - 5t^2 + 1,18$$

$$7,85t - v_f t - 1,18 = 0$$

$$(7,85 - v_f)t - 1,18 = 0$$

$$5t - 7,85t + 1,18 = 0$$

$$\frac{7,85 \pm \sqrt{61,62 - 23,6}}{10}$$

$$\frac{7,85 + 6,16}{10} = 1,4$$

$$\frac{7,85 - 6,16}{10} = 0,16 \rightarrow \text{più plausibile } v = \frac{s}{t} \rightarrow t = \frac{s}{v}$$

Percorre 1,18 m in 0,16 s

$$v_f t - \frac{1}{2} g t^2 = 1,18$$

$$v_f(0,16) - 5(0,16)^2 = 1,18$$

$$v_f(0,16) - 0,128 = 1,18$$

$$v_f = 8,17$$

In generale, a è costante:

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$v = v_0 + at$$

$$v_f = 7,85 - 10(0,16)$$

$$v_f = 7,85 - 1,6 = 6,25$$

$$\Delta E_m = 0$$

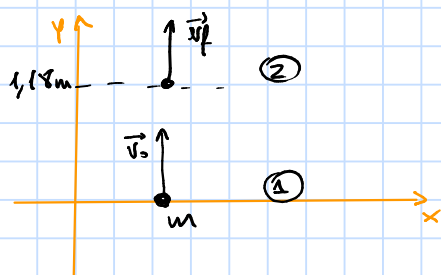
$$E_m(1) = E_m(2)$$

$$\frac{mv^2}{2} = \frac{mv_0^2}{2} + mgh$$

$$\frac{(0,120)(7,85)^2}{2} = \frac{(0,120)v_f^2}{2} + (0,120)(10)(1,18)$$

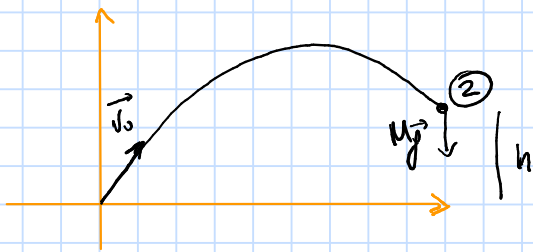
$$3,7 = 0,06 v_f^2 + 1,416$$

$$\sqrt{\frac{2,284}{0,06}} = v_f \rightarrow v_f = 6,21$$



L'unica forza che fa lavoro è la forza peso ed è una forza conservativa

2)



$$v_0 = 36 \text{ m/s}$$

$$m = 0,15 \text{ kg}$$

$$h = 7,2 \text{ m}$$

$$E_c(2) = ?$$

$$\Delta E_m = 0$$

$$E_m(1) = E_m(2)$$

$$\frac{mv_0^2}{2} = \frac{mv_f^2}{2} + mgh \rightarrow \frac{0,15 \cdot 36^2}{2} = \frac{0,15(v_f^2)}{2} + 0,15(10)(7,2)$$

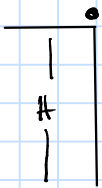
$$E_c(2) = \frac{mv_f^2}{2}$$

$$97,2 = 0,075 v_f^2 + 10,8$$

$$E_c(2) = 86,7 \text{ J}$$

$$\frac{86,4}{0,075} = v_f^2 \quad v_f = 34 \text{ m/s}$$

3)



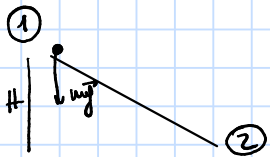
$$h = 60 \text{ m}$$

$$-\Delta E_p = 25000 \text{ J}$$

$$-\Delta E_p = E_p(1) - E_p(2) = mgh - 0 \rightarrow 25000 = m(10)(60) \rightarrow m = 625 \text{ kg}$$

$$mg = 625 \text{ N} \quad \text{con} [N] = \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

4)



$$h = 2,61 \text{ m}$$

$$v_0 = 0$$

$$v_f = ?$$

$$\Delta E_m = 0$$

$$E_c(1) + E_p(1) = E_c(2) + E_p(2)$$

$$mg \cdot 2,61 = \frac{1}{2} m v_f^2$$

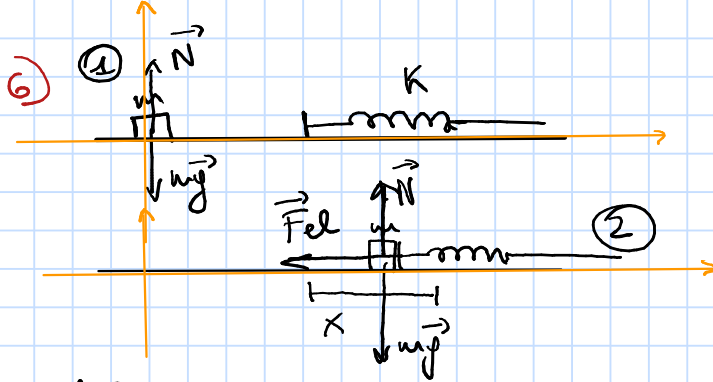
$$26,1 = \frac{1}{2} v_f^2 \rightarrow v_f = 7,22 \text{ m/s}$$

5)

$$v_0 = 0,840 \text{ m/s}$$

$$mg \cdot 2,61 + \frac{m \cdot 0,840^2}{2} = \frac{1}{2} m v_f^2$$

$$26,80 = \frac{1}{2} v_f^2 \rightarrow v_f = 7,32 \text{ m/s}$$



$$m = 1,70 \text{ kg}$$

$$K = 955 \text{ N/m}$$

$$x = 6,60$$

$$v_i = ?$$

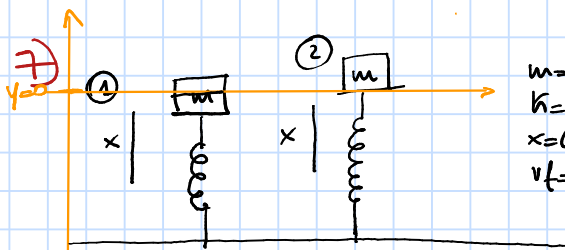
$$\Delta E_m = 0$$

$$E_p(1) + E_c(1) = E_p(2) + E_c(2)$$

$$\frac{mv_i^2}{2} = \frac{Kx^2}{2}$$

$$\frac{1,70}{2} v_i^2 = \frac{955 (6,60)^2}{2}$$

$$985 v_i^2 = 401 \rightarrow v_i = 4,09 \text{ m/s}$$



$$m = 1,70$$

$$K = 955 \text{ N/m}$$

$$x = 6,60$$

$$v_f = ?$$

$$\Delta E_m = 0$$

$$E_m(1) = E_m(2)$$

$$mgh + \frac{Kx^2}{2} = \frac{mv_f^2}{2}$$

$$1,7(-0,0660) + \frac{955(-0,0660)^2}{2} = \frac{1,7}{2} v_f^2$$

$$-0,782 + 4,01 = 0,85 v_f^2$$

$$3,22 = v_f^2 \rightarrow v_f = 1,79 \text{ m/s}$$